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"MW-Gui-Builder" package: A graphical interface to create electrode/electrolyte interfaces for Metalwalls package application

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"MW-Gui-Builder": An interface for simplified preparation for Metalwalls package application

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Summary

MW-Gui-Builder is an open-source Python package with a graphical interface for preparing simulations using the MetalWalls molecular dynamics engine. It streamlines the creation of structure and parameter files for electrochemical systems, including electrode construction and ion packing, making it accessible for both new and advanced users. MW-Gui-Builder reduces the barrier to entry for accurate constant-potential simulations.

Statement of need

Molecular dynamics simulations have been revealed as powerfull, versatile tools for investigating chemical and biological systems without the limitations of experiments (Hollingsworth & Dror, 2018).
However, preparing systems for such simulations can be tedious and complex, especially for new users. The development of modern MD packages enables increasingly realistic and complete system descriptions, but also requires adapting the setup workflow to match current scientific challenges (Zeng et al., 2023).

The MetalWalls software, released in 2020, rapidly demonstrated its value in electrochemical simulations and now plays a major role in the field (Coretti et al., 2022; Scalfi et al., 2021). Its constant-potential method overcomes limitations of previous approaches, enabling accurate

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representation of electrochemical interfaces. Nevertheless, the presence of explicit metallic electrodes introduces new challenges in system construction. The specific file formats required by MetalWalls make earlier builders incompatible and the Metalwalls community specifically asked for a usable builder.

This paper introduces *MW-Gui-Builder*, a dedicated tool for building constant-potential simulations in MetalWalls. It generates structure and parameter files in the required formats, operates within a Python environment for extensibility, and provides a graphical user interface that simplifies simulation setup.

State of the field

A key step in running MetalWalls simulations is the generation of structure and input files that comply with the MetalWalls format, which encodes atomistic geometries, electrode definitions, and simulation parameters. While this format is powerful and flexible, manually creating or modifying these files can be time-consuming and error-prone, especially for complex geometries or for users less familiar with the internal structure of MetalWalls inputs. The most usual way consist in the use of a bulk creation software (like Packmol used in this case), and the addition of a structured electrode. However no codes can today adresses the correct creation of the parameter file specific to the MetalWalls software.

MW-Gui-Builder addresses this limitation by providing a dedicated graphical user interface (GUI) for the construction of MetalWalls-compatible structure files. The GUI enables users to visually assemble and manipulate atomic systems, significantly lowering the barrier to entry for setting up simulations. A central design choice of MW-Gui-Builder is its reliance on an adaptable `.xyz` format to describe atomic coordinates. The `.xyz` format is lightweight, human-readable, and widely supported by atomistic simulation and visualization tools, which facilitates interoperability and rapid prototyping. Within MW-Gui-Builder, `.xyz` files can be easily imported, edited, and extended to include MetalWalls-specific information, allowing users to iteratively refine their systems through an intuitive graphical workflow. By combining a user-friendly GUI with the flexibility of the `.xyz` format, MW-Gui-Builder streamlines the creation of MetalWalls input files and promotes reproducible and accessible simulation setup practices.

Software design

MW-Gui-Builder is organised around a clear and modular architecture that separates user interaction from scientific data processing. This design choice makes the software easy to use for beginners while remaining maintainable and extensible for developers.

The application is launched from a single entry point that initializes the graphical user interface (GUI). The GUI acts as the central control panel, allowing users to access the different tools required to prepare MetalWalls simulations. Each tool corresponds to a well-defined task in the simulation workflow, such as structure conversion, electrode generation, or input file preparation.

Behind the interface, the core logic is implemented in dedicated Python modules. These modules are responsible for reading molecular structure files, processing atomic and electrode data, enforcing the expected ordering and conventions, and writing simulation-ready input files. The GUI only handles user input, validation, and feedback, while all scientific and file-related operations are delegated to these core modules.

This separation ensures a clean data flow: user actions trigger GUI callbacks, which call the appropriate core functions and return results or error messages to the interface. By following standard molecular dynamics file formats and keeping responsibilities clearly separated, MW-Gui-Builder integrates smoothly into existing workflows and can be easily extended with new features or tools.

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Research Impact Statement

This software has been developped during a PhD project and was used to produce several initial system that have been published in ACS Omega in 2026

Example

All examples are described in detail in the tutorials available on the project's GitHub repository. The molecules and electrode files used in each example are located in the `molecules/` directory.

Water between graphene

The structure shown in **Figure 1** is constructed from 5000 TIP4P/2005 water molecules packed in a cubic arrangement. Two graphene electrodes are placed at $z = 0$ *Å* and $z = 58$ *Å*. A packing buffer of *2 Å* is used, and the desired electrode thickness is *8 Å*. The software automatically adjusts electrode thickness to the closest lower value (here *6.7 Å*).

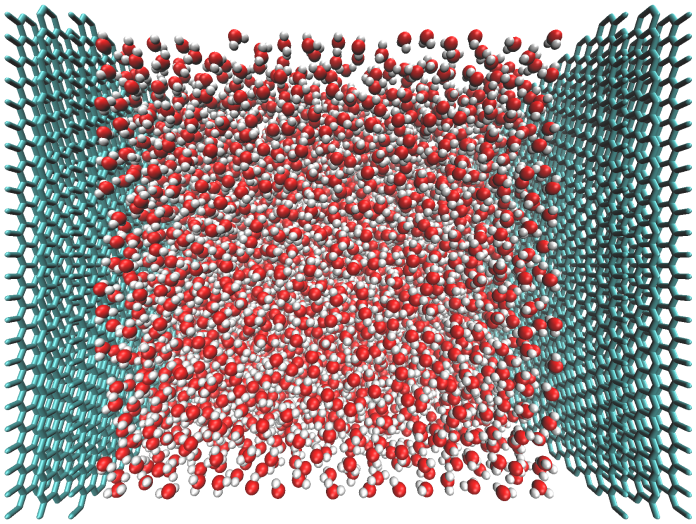


Figure 1: Image 1

Figure 1: Cubic water liquid between two graphene electrodes.

Ionic liquid with a central separator

Electrodes can also be placed in the center of the simulation cell using full 3D periodicity. In **Figure 2**, 400 pairs of [EMIM][TFSI] ionic liquid are packed at a concentration of 5 M/L. The box is twice as large in the z-direction, and a Pt(110) electrode is placed at $z = 65$ *Å* with a buffer of *7 Å* and a target length of *8 Å*.

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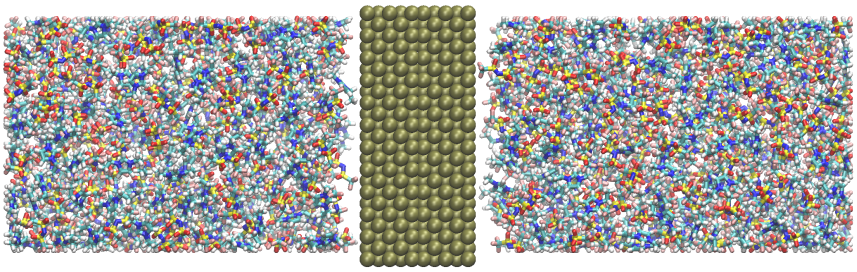


Figure 2: Image 2

Figure 2: Pt(110) electrode dividing a box of [EMIM][TFSI] ionic liquid.

Electrochemically meaningful system

The system in **Figure 3** contains 3000 TIP4P water molecules and 54 NaOH ion pairs, corresponding to a 1 M electrolyte.
Two Pt(100) electrodes of length *8 Å* are placed at $z = 0$ *Å* and $z = 130$ *Å*.

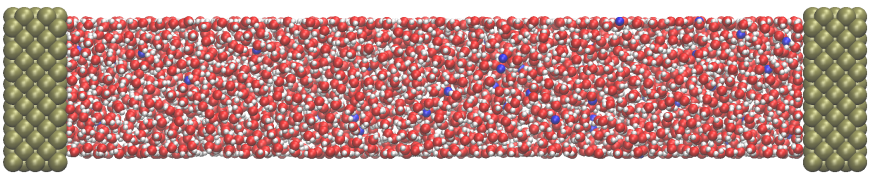


Figure 3: Image 3

Figure 3: 1 M NaOH in TIP4P water between two Pt(100) electrodes.

Related projects

The package is currently in active use at the *Institut de Chimie et Procédés pour l'Énergie, l'Environnement et la Santé (ICPEES)* as part of multiple research initiatives focusing on the electrochemical double layer. These projects fall within the DECODE project (Horizon Europe 101135537), which aims to elucidate local reaction environments in electrochemical systems.

This package was also included in the FULLMAP project (Horizon 101192848) to provide base into material simulation process.

Future developments will extend MW-Gui-Builder to support more complex electrodes, such as curved surfaces, partially covered electrodes, and bimetallic materials. These additions will allow researchers to model a broader range of realistic electrochemical interfaces.

AI usage disclosure

No generative AI tools were used in the development of this software, the writing of this manuscript, or the preparation of supporting materials.

References

Coretti, A., Bacon, C., Berthin, R., Serva, A., Scalfi, L., Chubak, I., Goloviznina, K., Haefele, M., Marin-Laflèche, A., Rotenberg, B., Bonella, S., & Salanne, M. (2022). MetalWalls: Simulating electrochemical interfaces between polarizable electrolytes and metallic electrodes. *The Journal of Chemical Physics*, *157*(18), 184801. https://doi.org/10.1063/5.0101777

Hollingsworth, S. A., & Dror, R. O. (2018). Molecular dynamics simulation for all. *Neuron*, *99*(6), 1129–1143. https://doi.org/https://doi.org/10.1016/j.neuron.2018.08.011

Scalfi, L., Salanne, M., & Rotenberg, B. (2021). Molecular simulation of electrode-solution interfaces [Journal Article]. *Annual Review of Physical Chemistry*, *72*(Volume 72, 2021), 189–212. https://doi.org/https://doi.org/10.1146/annurev-physchem-090519-024042

Zeng, L., Peng, J., Zhang, J., Tan, X., Ji, X., Li, S., & Feng, G. (2023). Molecular dynamics simulations of electrochemical interfaces. *The Journal of Chemical Physics*, *159*(9), 091001. https://doi.org/10.1063/5.0160729

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